LECTURE 15 (153-248)

SMOOTHING THE PERIODOGRAM

Time Series: $y_0 \dots y_{n-1}$

DFT: $b_0 \dots b_{n-1}$

$$b_j = \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i j t}{n}\right)$$

Periodogram: $I\left(\frac{j}{n}\right) = \frac{|b_j|^2}{n}$

$j = 1, \dots, m$

n odd
$m = \frac{n-1}{2}$

Fact: $b_{n-j} = \overline{b_j}$

$$RSS\left(\frac{j}{n}\right) = \min_{\beta,\beta_1,\beta_2} \sum_t \left(y_t - \beta_0 - \beta_1 \cos 2\pi \frac{j}{n} t - \beta_2 \sin \frac{2\pi j}{n} t\right)^2$$

$$= \sum (y_t - \bar{y})^2 - 2 I\left(\frac{j}{n}\right)$$

Related Fact: n odd $\left(m = \frac{n-1}{2}\right)$

$$2 I\left(\frac{1}{n}\right) + 2 I\left(\frac{2}{n}\right) + \cdots + 2 I\left(\frac{m}{n}\right) = \sum_t (y_t - \bar{y})^2$$

$$RSS\left(\frac{j_1}{n}, \frac{j_2}{n}\right) = \sum (y_t - \bar{y})^2 - 2 I\left(\frac{j_1}{n}\right) - 2 I\left(\frac{j_2}{n}\right)$$

In many applications, periodogram is noisy (with noise more apparent on the log-scale)

We are looking at a noise removal technique for the periodogram. leading to a smooth structure

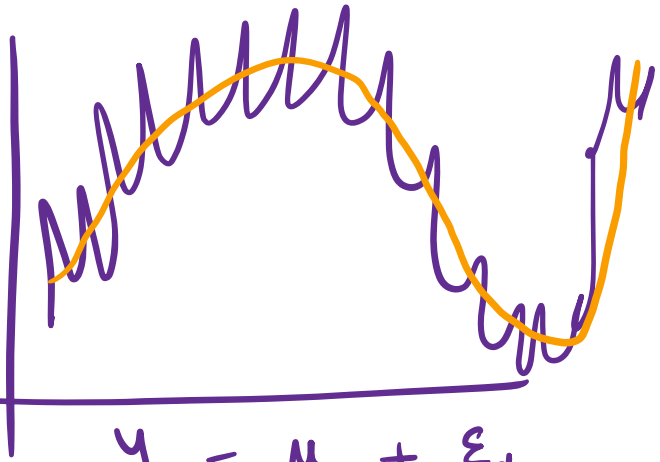
$$I\left(\frac{j}{n}\right) = f\left(\frac{j}{n}\right) + \varepsilon_j$$

$\varepsilon_j \overset{iid}{\sim} N(0,\sigma^2)$

μ_j

$$\log I\left(\frac{j}{n}\right) = \log f\left(\frac{j}{n}\right) + \varepsilon_j$$

$\varepsilon_j \overset{iid}{\sim} N(0,\sigma^2)$



$$y_t = \mu_t + \varepsilon_t$$

$y_1 \cdots y_n$

μ_t: clean smooth signal

$\varepsilon_t \overset{iid}{\sim} N(0,\sigma^2)$

$$\sum_{t=1}^{n} (y_t - \mu_t)^2 + \lambda \sum_t \left[(\mu_t - \mu_{t-1}) - (\mu_{t-1} - \mu_{t-2})\right]^2$$

$$\sum_{t=3}^{n} (\mu_t - 2\mu_{t-1} + \mu_{t-2})$$

Model we will use:

$$I\left(\frac{j}{n}\right) = f\left(\frac{j}{n}\right) \times \eta_j$$

$f\left(\frac{j}{n}\right)$: mean of $I(j/n)$

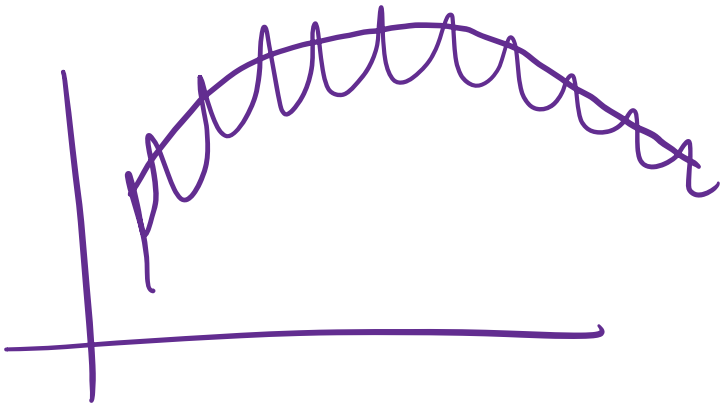
η_j: multiplicative noise with mean 1

$$\eta_j \overset{iid}{\sim} \frac{\chi^2_2}{2}$$

$$I\left(\frac{j}{n}\right) = \frac{|b_j|^2}{n}$$

Alternatively,

$$\log I\left(\frac{j}{n}\right) = \log f\left(\frac{j}{n}\right) + \log \eta_j$$



$$\eta_j \overset{iid}{\sim} \frac{\chi^2_2}{2}$$

$$\chi^2_K = \text{Gamma}\left(\frac{K}{2}, \frac{1}{2}\right)$$

density: $\propto x^{\frac{K}{2}-1} e^{-\frac{1}{2}x}$ $I(x>0)$

$K=2$ density of $e^{-\frac{1}{2}x} I(x>0)$

$$\chi^2_2 = \text{Exponential}\left(\frac{1}{2}\right)$$

$$\frac{\chi^2_2}{2} = \frac{1}{2}\,\text{Exp}\left(\frac{1}{2}\right) = \text{Exp}(1)$$

Model:

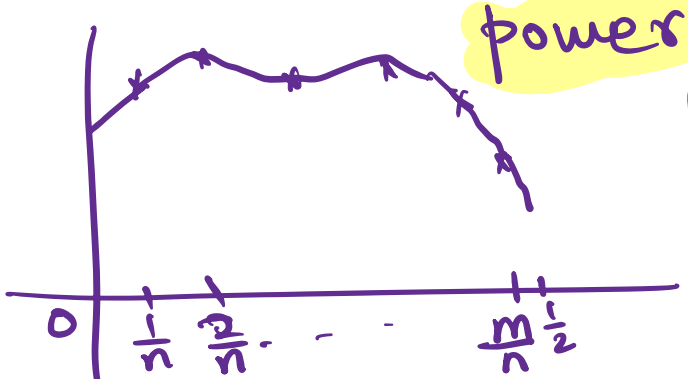
$$I\left(\frac{j}{n}\right) = f\left(\frac{j}{n}\right)\eta_j \qquad \eta_j \overset{iid}{\sim} \text{Exp}(1)$$

$\rightarrow$ Spectrum Model.

Goal: To estimate $f\left(\frac{j}{n}\right)$

power of frequency $\frac{j}{n}$

POWER SPECTRAL DENSITY



LAST LECTURE

DOES NOT INTEGRATE TO 1.

$$\mathrm{Re}(b_j), \mathrm{Im}(b_j) \overset{iid}{\sim} N(0, \gamma_j^2), \quad j = 1, \ldots, m$$

$$\Downarrow$$

$$I\left(\frac{j}{n}\right) = \frac{(\mathrm{Re}(b_j))^2 + (\mathrm{Im}(b_j))^2}{n}$$

$$= \frac{\gamma_j^2}{n} \left(\text{Sum of squares of 2 standard normals}\right)$$

$$= \frac{2\gamma_j^2}{n} \frac{\chi_2^2}{2}$$

$$= f\left(\frac{j}{n}\right) \frac{\chi_2^2}{2} \quad \text{where } f\left(\frac{j}{n}\right) = \frac{2\gamma_j^2}{n}$$

Estimation

$$I\left(\frac{j}{n}\right) = f\left(\frac{j}{n}\right) \eta_j, \quad \eta_j \overset{iid}{\sim} \mathrm{Exp}(1)$$

$$\overset{ind}{\sim} \mathrm{Exp}\left(\text{mean } f\left(\frac{j}{n}\right)\right)$$

Likelihood:

$$\prod_{j=1}^{m} \frac{1}{f(j/n)} \exp\left(-\frac{I(j/n)}{f(j/n)}\right)$$

$\mathrm{Exp}(\lambda)$: $\lambda e^{-\lambda x}$

$\mathrm{Exp}(\text{mean } \mu)$: $\frac{1}{\mu} e^{-\frac{x}{\mu}} I(x>0)$

Log-likelihood.

$$\sum_{j=1}^{m}\left[-\frac{I(j/n)}{f(j/n)}-\log f(j/n)\right]$$

Negative log-likelihood:

$$\sum_{j=1}^{m}\left[\frac{I(j/n)}{f(j/n)}+\log f(j/n)\right]$$

$$f\left(\frac{j}{n}\right)=\frac{2\gamma_j^2}{n}$$

$$\sum_{j=1}^{m}\left[\frac{n\,I(j/n)}{2\gamma_j^2}+2\log\gamma_j\right]+\text{constant}.$$

Switch optimization variable: $\alpha_j=\log\gamma_j$

$$\sum_{j=1}^{m}\left[\frac{n\,I(j/n)}{2}e^{-2\alpha_j}+2\alpha_j\right]$$

$\alpha_1\cdots\alpha_m$

Regularized Estimation

Minimize $\sum_{j=1}^{m}\left[\frac{n\,I(j/n)}{2}e^{-2\alpha_j}+2\alpha_j\right]$

$$+\lambda\sum_{j=2}^{m-1}\left(\alpha_{j+1}-2\alpha_j+\alpha_{j-1}\right)^2$$

$$\text{or } \lambda \sum_{j=2}^{n-1} |\alpha_{j+1} - 2\alpha_j + \alpha_{j-1}|$$

$\hat{\alpha}_j$

$$\hat{f}\left(\frac{j}{n}\right) = \frac{2\gamma_j^2}{n} = \frac{2e^{2\hat{\alpha}_j}}{n}$$

FACT: DFT has equivalent information to the original data.

$y_0 \ldots y_{n-1} \xrightarrow{DFT} b_0 \ldots b_{n-1}$

$$b_j = \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i j t}{n}\right)$$

Data
Inverse DFT

$$y_t = \frac{1}{n} \sum_{j=0}^{n-1} b_j \exp\left(+\frac{2\pi i j t}{n}\right)$$

From here, deduce

n odd
$m = \frac{n-1}{2}$

$$y_t = \frac{b_0}{n} + \sum_{j=1}^{m} \left[\frac{2\,\mathrm{Re}(b_j)}{n} \cos\frac{2\pi j t}{n} + \frac{-2\,\mathrm{Im}(b_j)}{n} \sin\frac{2\pi j t}{n} \right]$$

$$y_t = \beta_0 + \sum_{j=1}^{m} \left[\beta_{1j} \cos\frac{2\pi j t}{n} + \beta_{2j} \sin\frac{2\pi j t}{n} \right]$$

Model: $\mathrm{Re}(b_j), \mathrm{Im}(b_j) \overset{iid}{\sim} N(0, \gamma_j^2)$

$\beta_{1j}, \beta_{2j} \overset{iid}{\sim} N(0, \tau_j^2)$

$$y_t = \beta_0 + \sum_{j=1}^{m} \left[\beta_{1j} \cos \frac{2\pi j t}{n} + \beta_{2j} \sin \frac{2\pi j t}{n} \right]$$

$\beta_{1j}, \beta_{2j} \overset{iid}{\sim} N(0, \tau_j^2)$

$\downarrow$

Check:

$$\mathbb{E}\, y_t = \beta_0 \qquad \tau_j^2 = \frac{2}{n} f\left(\frac{j}{n}\right)$$

$$\mathrm{var}(y_t) = \sum_{j=1}^{m} \tau_j^2 = \sum_{j=1}^{m} \frac{2}{n} f\left(\frac{j}{n}\right)$$

$$\mathrm{Cov}(y_t, y_{t+h}) = \mathrm{Cov}\left(\sum_j \beta_{1j} \cos \frac{2\pi j t}{n} + \beta_{2j} \sin \frac{2\pi j t}{n}, \right.$$

$$\left. \sum_j \beta_{1j} \cos \frac{2\pi j (t+h)}{n} + \beta_{2j} \sin \frac{2\pi j (t+h)}{n} \right)$$

$$\mathrm{Cov}(y_t, y_{t+h}) = \sum_j \tau_j^2 \cos \frac{2\pi j h}{n}$$